# PE2E SEARCH - Search History (Prior Art)

| Ref # | Hits | Search Query | DBs | Default Operator | Plurals | British Equivalents | Time Stamp |
|---|---|---|---|---|---|---|---|
| L1 | 4 | ("10094691").pn. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 09:46 AM |
| L2 | 37 | (Artificial Intelligence)<br>More like doc: US-10094691-B2<br>Text: (EP-2930475-B1 OR US-20160216144-A1 OR EP-2930475-A1 OR EP-1840535-A1 OR EP-3404373-B1 OR EP-1840535-B1 OR JP-5112728-B2 OR JP-2007279036-A OR EP-3404373-A1 OR US-8011240-B2 OR US-20070241093-A1 OR JP-2022139173-A OR US-20220291028-A1 OR EP-4056961-A1 OR CN-115077641-A OR JP-20070279036-A5 OR EP-1840536-A1 OR US-20070227242-A1 OR US-7490511-B2 OR EP-1840536-B1).did. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 09:49 AM |
| L3 | 4 | "200215283" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, | OR | ON | ON | 2023/11/20 09:51 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L4 | 11 | "2002015283" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 09:51 AM |
| L5 | 9 | "2002016283" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 09:51 AM |
| L6 | 5 | ("6779712").pn. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:02 AM |
| L7 | 39 | (Artificial Intelligence)<br>More like doc:<br>US-6779712-B2<br>Text: | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, | OR | ON | ON | 2023/11/20 10:05 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | (US-20040000196-A1 OR EP-1351039-A1 OR DE-10191688-B4 OR US-7408133-B2 OR US-20070044554-A1 OR EP-1917503-A1 OR WO-2007025001-A1 OR CN-113097200-A OR CN-101292135-B OR CN-101292135-A OR DE-102018109971-A1 OR WO-2019120901-A1 OR JP-2836027-B2 OR AU-4868901-A OR AT-E301822-T1 OR US-11092471-B2 OR CN-109307536-A OR US-20220291028-A1 OR EP-4056961-A1 OR JP-2022139173-A).did. | EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L8 | 7 | "20140102086" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:21 AM |
| L9 | 15 | "2014102086" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:22 AM |
| L10 | 29 | "2011030468" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, | OR | ON | ON | 2023/11/20 10:22 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L11 | 13 | "20110030468" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:22 AM |
| L12 | 3 | "20120216629" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:23 AM |
| L13 | 5 | "20140190252" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, | OR | ON | ON | 2023/11/20 10:23 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L14 | 3 | "20090158859" | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:23 AM |
| L15 | 21 | ("4548078" \| "4829818" \| "5081866" \| "5228329" \| "5705745" \| "6187609" \| "6351390" \| "6443179" \| "6502459" \| "6548895" \| "6600202" \| "6617963" \| "20020190839" \| "20030098771" \| "20040000196" \| "20040021185" \| "20040050155").pn. OR ("6779712").urpn. AND (PGPB \| USPT \| USOC).dbnm. | (US-PGPUB; USPAT; USOCR) | OR | ON | ON | 2023/11/20 10:26 AM |
| L16 | 15840 | (G01F1/6888 OR G01N25/18 OR G01F1/688 OR G01F15/16).cpc. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:28 AM |
| L17 | 62236986 | (metal steel aluminum titanium copper tungsten molybdenum tantalum) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, | OR | ON | ON | 2023/11/20 10:31 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L18 | 116855231 | (cavity OR recess OR recesses OR recessed OR hole OR chamber$2 OR ante$1chamber$2 OR hollow OR hollowed OR pocket OR pocketed OR space OR sub$1space$1 OR sub$1spacing OR void OR gap OR micro$2gap$1 OR nano$1gap$1 OR cell OR alcove OR alcoved OR burrow OR cave OR hollow$4 OR vestibul$4 OR inlay OR inlaid OR inlayed OR inlaying2 OR inter$space$1) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:31 AM |
| L19 | 85085667 | (hole OR through$1hole$1 OR pin$1hole$1 OR perforate OR perforated OR perforation OR aperture OR orifice OR sink$1hole$1 OR bore OR bored OR boring OR bore$1hole$1 OR well$1bore$1 OR cubby$1hole$1 OR port OR port$1hole$1 OR guide$1hole$1 OR through$1via$1 OR (through ADJ via) OR inlet OR outlet) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:31 AM |
| L20 | 99680700 | (aperture OR orifice OR opening OR hole OR through$1hole$1 OR pin$hole$1 OR port OR port$1hole$1 OR through$1port$1 OR guide$1hole$1 OR inlet OR outlet OR through$1via$1 OR (through ADJ via) ) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; | OR | ON | ON | 2023/11/20 10:32 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | JPO; DERWENT; IBM_TDB) | | | | |
| L21 | 42702016 | (port OR port$1hole$1 OR through$1port$1 OR inlet OR outlet) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:32 AM |
| L22 | 56 | (EP-4056961-$ JP-2022139173 ADJ2 (A$1 OR B$1) CN-115077641-$ WO-2020158155-$ JP-2020122747 ADJ2 (A$1 OR B$1) JP-2008304293 ADJ2 (A$1 OR B$1) JP-5224089 ADJ2 (A$1 OR B$1) JP-7258800 ADJ2 (A$1 OR B$1) WO-2022030039-$ JP-2001318106 ADJ2 (A$1 OR B$1) JP-2002054963 ADJ2 (A$1 OR B$1) JP-7258797 ADJ2 (A$1 OR B$1) US-20230101259-$ JP-2002005714 ADJ2 (A$1 OR B$1) CN-115280113-$ JP-2002168876 ADJ2 (A$1 OR B$1) JP-2014160030 ADJ2 (A$1 OR B$1) JP-3710994 ADJ2 (A$1 OR B$1) JP-3589083 ADJ2 (A$1 OR B$1) JP-2022030781 ADJ2 (A$1 OR B$1) WO-2012111368-$ JP-5874117 ADJ2 (A$1 OR B$1) JP-2000292234 ADJ2 (A$1 OR B$1) JP-2002162271 ADJ2 (A$1 OR B$1) CN-115151794-$).did. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:34 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| L23 | 218 | (EP-4056961-$ JP-2022139173 ADJ2 (A$1 OR B$1) CN-115077641-$ WO-2020158155-$ JP-2020122747 ADJ2 (A$1 OR B$1) JP-2008304293 ADJ2 (A$1 OR B$1) JP-5224089 ADJ2 (A$1 OR B$1) JP-7258800 ADJ2 (A$1 OR B$1) WO-2022030039-$ JP-2001318106 ADJ2 (A$1 OR B$1) JP-2002054963 ADJ2 (A$1 OR B$1) JP-7258797 ADJ2 (A$1 OR B$1) US-20230101259-$ JP-2002005714 ADJ2 (A$1 OR B$1) CN-115280113-$ JP-2002168876 ADJ2 (A$1 OR B$1) JP-2014160030 ADJ2 (A$1 OR B$1) JP-3710994 ADJ2 (A$1 OR B$1) JP-3589083 ADJ2 (A$1 OR B$1) JP-2022030781 ADJ2 (A$1 OR B$1) WO-2012111368-$ JP-5874117 ADJ2 (A$1 OR B$1) JP-2000292234 ADJ2 (A$1 OR B$1) JP-2002162271 ADJ2 (A$1 OR B$1) CN-115151794-$ KR-20090052173-$ JP-3710995 ADJ2 (A$1 OR B$1) KR-100931702-$ WO-2021176793-$ JP-2012111368 ADJ2 (A$1 OR B$1) JP-2021139652 ADJ2 (A$1 OR B$1) US-20230113770-$ JP-2000074713 ADJ2 (A$1 OR B$1) JP-4709499 ADJ2 (A$1 OR B$1) WO-2021166401-$ CN-114902020-$ US-10094691-$ JP-2019138780 ADJ2 (A$1 OR B$1) EP-2930475-$ EP-2677283-$ JP- | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:34 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 7112373 ADJ2 (A$1 OR B$1) WO-2020158156-$ US-20120318058-$ US-8667839-$ TW-202134605-$ JP-2004340964 ADJ2 (A$1 OR B$1) JP-2021135053 ADJ2 (A$1 OR B$1) JP-3252375 ADJ2 (A$1 OR B$1) US-8943888-$ US-20140190251-$ JP-6685789 ADJ2 (A$1 OR B$1) JP-$1"09257821"- US-6945106-$ JP-2001249040 ADJ2 (A$1 OR B$1) JP-2002139510 ADJ2 (A$1 OR B$1) US-20160216144-$ JP-2017173126 ADJ2 (A$1 OR B$1) US-20100078753-$ WO-2016132935-$ US-20210063248-$ EP-1477781-$ JP-2002286519 ADJ2 (A$1 OR B$1) US-20200300711-$ JP-2007147429 ADJ2 (A$1 OR B$1) US-11454553-$ JP-2020122752 ADJ2 (A$1 OR B$1) JP-2016151473 ADJ2 (A$1 OR B$1) JP-2002202168 ADJ2 (A$1 OR B$1) EP-1477781-$ US-20040261520-$ WO-2016132934-$ KR-20000033148-$ JP-5043556 ADJ2 (A$1 OR B$1) US-10876903-$ KR-100292799-$ JP-2008233057 ADJ2 (A$1 OR B$1) JP-2009042097 ADJ2 (A$1 OR B$1) EP-2930475-$ JP-2018141664 ADJ2 (A$1 OR B$1) JP-4995617 ADJ2 (A$1 OR B$1) JP-6467172 ADJ2 (A$1 OR B$1) JP-3658321 ADJ2 (A$1 OR B$1) JP-2000310553 ADJ2 (A$1 OR B$1) US- | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 20210116281-$ KR-20010019419-$ JP-2016109527 ADJ2 (A$1 OR B$1) JP-3530069 ADJ2 (A$1 OR B$1) JP-$1"1164061"- JP-2004061983 ADJ2 (A$1 OR B$1) EP-4048983-$ JP-2016151472 ADJ2 (A$1 OR B$1) ES-2347145-$ JP-2020193805 ADJ2 (A$1 OR B$1) JP-3381831 ADJ2 (A$1 OR B$1) KR-100362010-$ JP-2018115976 ADJ2 (A$1 OR B$1) US-20170343422-$ JP-2016061592 ADJ2 (A$1 OR B$1) CN-113677962-$ US-20220333966-$).did. | | | | | |
| L24 | 162 | L23 NOT L22 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:45 AM |
| L25 | 2 | ("20020121137").pn. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 10:54 AM |
| L26 | 7561132 | L17 near4 (L18 L19 L20 L21 ) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, | OR | ON | ON | 2023/11/20 11:12 AM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L27 | 10007462 | L17 near6 (L18 L19 L20 L21 ) | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 11:12 AM |
| L28 | 1175 | L27 AND L16 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 11:14 AM |
| L29 | 39 | (Artificial Intelligence) More like doc: US-6779712-B2<br>Text: (US-20040000196-A1 OR EP-1351039-A1 OR DE-10191688-B4 OR US-7408133-B2 OR US-20070044554-A1 OR EP-1917503-A1 OR WO-2007025001-A1 OR CN-113097200-A | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, | OR | ON | ON | 2023/11/20 12:14 PM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | OR CN-101292135-B OR CN-101292135-A OR DE-102018109971-A1 OR WO-2019120901-A1 OR JP-2836027-B2 OR AU-4868901-A OR AT-E301822-T1 OR US-11092471-B2 OR CN-109307536-A OR US-20220291028-A1 OR EP-4056961-A1 OR JP-2022139173-A).did. | TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L30 | 37 | (Artificial Intelligence) More like doc: DE-10191688-B4<br>Text: (WO-0184099-A1 OR EP-1279008-B1 OR DE-20122320-U1 OR US-5705745-A OR KR-100515422-B1 OR DE-102019110876-A1 OR DE-102006035000-B4 OR DE-102006035000-A1 OR US-20040035201-A1 OR JP-2005531771-A OR US-6883370-B2 OR US-7021136-B2 OR EP-1499860-A4 OR EP-1499860-A1 OR US-20050087011-A1 OR KR-970007308-A OR JP-H07174599-A OR RU-2290610-C2 OR DE-2407374-A1 OR CN-1666089-A).did. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 12:16 PM |
| L31 | 39 | (Artificial Intelligence) More like doc: US-6779712-B2<br>Text: (US-20040000196-A1 OR EP-1351039-A1 OR DE-10191688-B4 OR US-7408133-B2 OR US-20070044554-A1 OR EP-1917503-A1 OR WO-2007025001-A1 OR CN-113097200-A OR CN-101292135-B OR CN-101292135-A OR DE-102018109971-A1 OR WO-2019120901-A1 OR JP-2836027-B2 OR AU-4868901-A OR AT- | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 12:16 PM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | E301822-T1 OR US-11092471-B2 OR CN-109307536-A OR US-20220291028-A1 OR EP-4056961-A1 OR JP-2022139173-A).did. | | | | | |
| L32 | 36 | (Artificial Intelligence) More like doc: US-6502459-B1<br>Text: (WO-0218884-A2 OR EP-1314042-A2 OR EP-1314042-B1 OR KR-20030064396-A OR KR-100825998-B1 OR JP-2004507753-A OR DE-60129992-T2 OR WO-0218884-A3 OR CN-101443635-A OR JP-2009529695-A OR EP-1994373-A2 OR WO-2007106689-A2 OR US-20070017285-A1 OR EP-2685223-A1 OR CN-103542908-A OR JP-2014016237-A OR US-7021136-B2 OR US-20040035201-A1 OR US-20050087011-A1 OR US-6883370-B2).did. | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 12:18 PM |
| L33 | 20755460 | (sold$5 weld$5) | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 01:01 PM |
| L34 | 187532 | (sold$5 weld$5) near5 ground$6 | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, | OR | ON | ON | 2023/11/20 01:12 PM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L35 | 1 | L34 AND L16 | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 01:12 PM |
| L36 | 5 | ((US-10094691-B2 OR US-6779712-B2 OR US-6502459-B1).did. AND USPT.dbnm.) OR ((US-20020121137-A1).did. AND PGPB.dbnm.) OR ((DE-10191688-B4).did. AND FTDB.dbnm.) | (US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 01:13 PM |
| L37 | 205019 | (G01F1/$).cpc. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 01:14 PM |
| L38 | 132 | L37 AND L34 | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, | OR | ON | ON | 2023/11/20 01:14 PM |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | | | | |
| L39 | 4 | US-4994035-$.did. \| US-5050429-$.did. | (US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB) | OR | ON | ON | 2023/11/20 02:33 PM |
| L40 | 99 | ("4733559").pn. OR ("5050429").urpn. AND (PGPB \| USPT \| USOC).dbnm. | (US-PGPUB; USPAT; USOCR) | OR | ON | ON | 2023/11/20 02:39 PM |

## PE2E SEARCH - Search History (Interference)

| Ref # | Hits | Search Query | DBs | Default Operator | Plurals | British Equivalents | Time Stamp |
|---|---|---|---|---|---|---|---|
| N1 | 969 | (G01F1/6888 OR G01N25/18 OR G01F1/688 OR G01F15/16).cpc. | (US-PGPUB) | OR | ON | ON | 2023/11/27 02:55 PM |
| N2 | 559525 | (port OR port$1hole$1 OR through$1port$1 OR inlet OR outlet).clm. | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:12 PM |
| N3 | 1685298 | (aperture OR orifice OR opening OR hole OR through$1hole$1 OR pin$hole$1 OR port OR port$1hole$1 OR through$1port$1 OR guide$1hole$1 OR inlet OR outlet OR through$1via$1 OR (through ADJ via) ).clm. | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:12 PM |
| N4 | 1299863 | (hole OR | (US-PGPUB) | OR | ON | ON | 2023/11/27 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | through$1hole$1 OR pin$1hole$1 OR perforate OR perforated OR perforation OR aperture OR orifice OR sink$1hole$1 OR bore OR bored OR boring OR bore$1hole$1 OR well$1bore$1 OR cubby$1hole$1 OR port OR port$1hole$1 OR guide$1hole$1 OR through$1via$1 OR (through ADJ via) OR inlet OR outlet).clm. | | | | | 03:13 PM |
| N5 | 2316393 | (cavity OR recess OR recesses OR recessed OR hole OR chamber$2 OR ante$1chamber$2 OR hollow OR hollowed OR pocket OR pocketed OR space OR sub$1space$1 OR sub$1spacing OR void OR gap OR micro$2gap$1 OR nano$1gap$1 OR cell OR alcove OR alcoved OR burrow OR cave OR hollow$4 OR vestibul$4 OR inlay OR inlaid OR inlayed OR inlaying2 OR inter$space$1).clm. | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:13 PM |
| N6 | 938699 | (metal steel aluminum titanium copper tungsten molybdenum tantalum).clm. | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:13 PM |
| N7 | 154459 | (sold$5 weld$5).clm. | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:13 PM |
| N8 | 102379 | (N2 N3 N4 N5 ) near5 (N6 N7 ) | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:13 PM |
| N9 | 26 | N8 AND N1 | (US-PGPUB) | OR | ON | ON | 2023/11/27 03:14 PM |